The Axiom of Choice: To Choose, or Not To Choose?

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Abstract

This essay will detail the history and development of the Axiom of Choice. As well as where the Axiom of Choice is used and why it was created in the first place. It will also discuss the Banach-Tarski paradox and how it fuels debates surrounding the validity of the Axiom of Choice.

1 Introduction

Axioms are the fundamental building blocks of all mathematics. They are the assumptions that must be made so that we can use them as basic facts in proofs. Set theory has axioms of its own, one of which being the *Axiom of Extensionality*, which states that sets are equal when they contain the exact same elements. We need axioms, because they define how the mathematical system we are working in functions. The axioms that are the basis of set theory mathematicians use today is called *Zermelo-Fraenkel Set Theory*, named after mathematicians Ernst Zermelo and Abraham Fraenkel. When mathematicians use *Zermelo-Fraenkel Set Theory*, they either include the *Axiom of Choice* or don't. We will see why this distinction is made through the course of this essay.

Definition 1.1. *Zermelo-Fraenkel Set Theory.* There are two formulations for this theory of sets. <u>ZF</u> denotes the system that excludes the axiom of choice, while <u>ZFC</u> denotes the system that includes the axiom of choice.

When the axiom of choice is combined with the axioms of ZF, we get ZFC. There are different equivalent statements for the AoC, we will use the definition that involves choice functions.

Definition 1.2. *Axiom of Choice (AoC).* For any collection of non-empty sets C, there exists a choice function f such that for any set S in C, f assigns an element of S to S. i.e $f(S) = x$ where $x \in S$ and S is any set in the collection C.

Example 1.3. Take the collection of sets $\{1,\ 2,\ 3\}$ $\{4,\ 5\}$ $\{99\}$. Let us define the function f so that $f(\{1, 2, 3\}) = 2$, $f(\{4, 5\}) = 5$, $f(\{99\}) = 99$. Thus, f is indeed a choice function for the collection.

The AoC tells us that for any collection of sets – where the collection and/or the sets are finite or infinite – there will always exist some choice function for that collection. That is, we can always find a way to choose an element from each set in the collection.

Now, one may wonder why we have both ZF and ZFC. Well, it is because the AoC is actually a controversial axiom that some mathematicians do not accept. Yet, at first glance it doesn't seem that the AoC is paradoxical or illogical, the problems with the AoC are only evident through closer examination. Before understanding the implications of this axiom and why it is controversial, we will first explore its beginnings.

2 A Well-Ordered Problem

In 1883, the German mathematician Georg Cantor surmised that within a collection of elements, there must be some method of ordering them. That is, after counting a first element, there exists a unique next element, as long as there are uncounted elements remaining. Cantor would title this type of ordering as "well-ordering". However, the verdict on whether every set could indeed be well-ordered - a *Well-Ordering Hypothesis*, so to speak - still remained undecided.

In 1900, David Hilbert - another German mathematician - would propose his famous 23 problems at the Paris conference of the International Congress of Mathematicians. One such question was the aforementioned Well-Ordering Hypothesis. Such a problem was noticed by a 29-year old mathematician, just six years after receiving his doctorate - Ernst Zermelo.

Ernst Friedrich Ferdinand Zermelo was a German Mathematician of the early 20th century. Born in 1871, Zermelo was encouraged to pursue academics from a young age - his father was a professor at a local school. In 1894, Zermelo would recieve his doctorate under Lazarus Immanuel Fuchs and Hermann Amandus Schwarz at the University of Berlin with his dissertation titled *Untersuchungen zur Variations-Rechnung* (Investigations in Calculus of Variations).

Following his doctorate, Zermelo studied hydrodynamics at the University of Berlin under Max Planck, the then-head of the physics department. Zermelo continued his research at the University of Göttingen after moving in 1897. He would publish his second dissertation in 1899, completing his habilitation and becoming a lecturer at the University of Göttingen.

Figure 2.1: Ernst Zermelo (1871-1953)

Zermelo showed interest in Hilbert's problems and began working on those that related to set theory. In 1904, Zermelo would prove that every set could be *well ordered*, leading to his appointment as a professor at Göttingham. Below is a similar proof to Zermelo's original solution to the Well-Ordering Problem:

Theorem 2.1. *Every set can be well-ordered.*

Proof. Let M be an arbitrary set of elements. Then, we denote its *power set*, or the set of all subsets of M as $P(M)$. Now, consider a choice function $\gamma : P(M) \setminus \{\emptyset\} \to M$ such that for some nonempty $m \in P(M)$, then $\gamma(m)$ returns an element from m.
We then define a γ-set M_γ as some set which satisfies the following conditions:

- $M_\gamma \subseteq M$.
- M_γ is well-ordered by some ordering ϕ.

- If $x \in M_\gamma$ is the $n-th$ element given our order ϕ, then we can create an initial segment $A = \{a_1, a_2, ..., a_{n-1}\}$, where $a_k \in M_\gamma$ is the $k-th$ element given ϕ. Now, $x = \gamma(M \setminus A)$.

Notice that such sets do exist. For example, given $m_1 = \gamma(M)$, we define the trivial well-ordering as $M_1 = \{m_1\}$. See that this is indeed a γ-set, as $m_1 \in M$, so $M_1 \subseteq M$. Moreover, there is only one element in $M1$, so the ordering ϕ_1 is trivial. Finally, the initial segment of M_1 given m_1 is the empty set, and $\gamma(M \setminus \emptyset) = \gamma(M) = m_1$, so all three conditions hold.

Let us recursively define the set $M_k = M_{k-1} \cup \{\gamma(M \setminus M_{k-1})\}$. Let us show that each set M_k is a γ-set through induction. Let M_k^c denote $M \setminus M_k$.
Base Case: We have shown above that $M_1 = \{\gamma(M)\}$ is a γ-set under the the ordering ϕ_1.
Induction Hypothesis: Assume that M_k is a γ-set under some well-ordering ϕ_k.
Inductive Step: We want to show that $M_{k+1} = M_k \cup \{m_{k+1}\}$, where $m_{k+1} = \gamma(M_k^c)$ is a γ-set.

- See that $m_{k+1} \in M$, and $M_k \subseteq M$, so $M_{k+1} \subseteq M$ as well.

- Now, let ϕ_{k+1} be an ordering for M_{k+1}, where the first k elements of M_{k+1} are ordered in the same way as they are for ϕ_k for M_k. Then, ϕ_{k+1} orders m_{k+1} as the "last" element.
  By the inductive hypothesis, ϕ_k is a well-ordering for M_k which we have extended by one element m_{k+1}. Thus, ϕ_{k+1} is also a well-ordering for $M_k \cup \{m_{k+1}\} = M_{k+1}$.

- Regarding the third condition, start with each element $m_j \in M_{k+1}$, where $1 \leq j \leq k$. Given the initial segment $A_j = \{m_1, \ldots, m_{j-1}\}$ by ϕ_{k+1}, consider $\gamma(M \setminus A_j)$. By the induction hypothesis, M_k is a γ-set under the well-ordering ϕ_k, whose ordering is identical to the first k elements of ϕ_{k+1}. Thus, we must have that $m_j = \gamma(M \setminus A_j)$, which holds for the first k elements.
  Next, for m_{k+1}, we have $A_{k+1} = \{m_1, \ldots, m_k\} = M_k$. However, by definition, we have that $m_{k+1} = \gamma(M_k^c\})$, so the third condition holds for the last element as well.
  Now, M_{k+1} satisfies all 3 conditions, so M_{k+1} is a γ-set as well.

Therefore, by the principle of mathematical induction, M_k is a γ-set for all $k \in \mathbb{N}$.

See that given the γ-sets M_k, M_{k+1}, we have that $M_{k+1} = M_k \cup \{\gamma(M_k^c)\}$. Since $\gamma(M_k^c)$ returns some element in M_k^c, and every element in M_k^c is not in M_k, we see that $|M_{k+1}| = |M_k| + 1$, and the extra element is distinct from all other elements of M_k. Therefore, M_{k+1} extends M_k, and we can say that the ordering ϕ_{k+1} extends the ordering ϕ_k as well.

Now, consider M_γ, which is the union over all the γ-sets. That is, M_γ is the largest γ-set which is a subset of M. In fact, M_γ must be M itself - otherwise, we can consider a larger γ-set $M'_\gamma = M_\gamma \cup \{\gamma(M_\gamma^c)\}$, which is a contradiction. However, as $M_\gamma = M$ is a γ-set, there must exist an ordering ϕ_γ which well-orders M, and so M must be well-ordered. $\square$

Formalizing such a proof required Zermelo to discover a new axiom of set theory: the **Axiom of Choice**. In fact, notice that the above proof invokes the axiom of choice when it implicitly declares the existence of a choice function γ.

3 Zermelo and Fraenkel's Set Theory

Zermelo would also develop an axiomatic set system in 1908, which included his newfound axiom of choice. **Zermelo Set Theory** is an axiomatic system built around the following axioms:

i) **Extensionality:** If every element in A is also an element of B, and every element of B is also an element of A, then A and B are equivalent. Each set is determined uniquely by its members.

ii) **Elementary Sets:**

- There exists a set with no elements $\emptyset$.
- For every element a in the domain, there exists a singleton set $\{a\}$.
- For all possible distinct pairs a, b in the domain, there exists the unordered pair $\{a, b\}$.

iii) **Seperation:** Given any set A and a condition z, we can create a set $A' \subseteq A$ which contains the elements of A which satisfy z.

iv) **Power Set:** For any set, the collection of all its subsets is also a set.

v) **Union:** For any set whose elements are also sets, then the union of the elements of the original set's elements is also a set.

vi) **Choice:** Given any collection of nonempty, pairwise disjoint sets, then there exists a set which contains exactly one member from each of the elements of the original set. (Notice that this definition of the axiom of choice is equivalent to the definition provided earlier in this paper, which includes a "choice function").

vii) **Infinity:** There exists some set of infinitely many elements. This set contains the empty set $\emptyset$, and for each element a, the set also contains the singleton set $\{a\}$.

However, Zermelo's original system was seen as flawed in the mathematical community. Many did not accept the Axiom of Choice, due to its vague nature. The choice function put forwards is never explicitly defined; the axiom simply states that such a function exists. The axiom of separation also came under scrutiny, as it was difficult to formulate the axiom into rigorous mathematical terms. In fact, each of the axioms stated above were only stated in plain English, and not in formal mathematical language.

In 1922, Abraham Fraenkel proposed a revised system of Zermelo's set theory, where each axiom is formally expressed in mathematical language. This new axiomatic system would come to be known as **Zermelo-Fraenkel set theory**, and consisted of the following axioms:

Figure 3.1: Abraham Fraenkel (1891-1965)

i) **Extensionality:** If two sets have the same the same members, they are the same set.

$$\forall A \forall B [\forall z \quad s.t. (z \in x \iff z \in y) \implies A = B]$$

ii) **Null Set:** There exists a set with no elements $\emptyset$.

$$\exists x \forall y \neg (y \in x)$$

iii) **Pairs:** There exists a unique pair set $\{x, y\}$ for each pair of elements x, y. If both elements are the same, then the set is simply a singleton set $\{x\}$.

$$\forall x \forall y \exists z \forall w (w \in z \iff w = x \vee w = y)$$

iv) **Power Set:** For every set x, there exists a set $P(x)$ which contains all of its subsets.

$$\forall x \exists y \forall z [z \in y \iff \forall w (w \in z \implies w \in x]$$

v) **Unions:** For every set x, there exists a set y which contains the elements of every element in x.

$$\forall x \exists y \forall z [z \in y \iff \exists w (w \in x \wedge z \in w)]$$

vi) **Infinity:** There exists some set x which contains the empty set. Also, for every set y which is an element in x, then the set containing every element of y and y itself is also in x.

$$\exists x [\emptyset \in x \wedge \forall y (y \in x \implies y \cup \{y\} \in x)]$$

vii) **Separation:** Given a set x and a collection of conditions $u_1, \ldots u_k$, then there exists a set y comprised of the elements in our original set x that satisfies all conditions.

$$\forall u_1 \ldots \forall u_k [\forall w \exists v \forall r (r \in v \iff r \in w \wedge \psi(r, u_1, \ldots, u_k))]$$

viii) **Replacement:** We call ϕ a functional formula if given a set x and a collection of conditions $u_1, \ldots u_k$, ϕ relates a unique set y to x under the given conditions. If a functional formula exists then given a collection of sets w, we can form a new set v which contains all elements which are uniquely related by ϕ to the elements of w under the conditions $u_1, \ldots u_k$.

$$\forall u_1 \ldots \forall u_k [\forall x \exists! y \phi(x, y, u_1, \ldots, u_k) \implies \forall w \exists v \forall r (r \in v \iff \exists s (s \in w \wedge \phi(s, r, u_1, \ldots u_k)))]$$

ix) **Regularity:** Every set contains a "minimal" element. This axiom is in place to prevent sets from becoming elements of themselves.

$$\forall x [x \neq \emptyset \implies \exists y (y \in x \wedge \forall z (z \in x \implies \neg (z \in y))]$$

x) **Choice*:** There exists some ordering for every set which well-orders it.

$$\forall x \exists \phi (\phi \text{ well-orders } x)$$

*This axiom is only present in the ZFC set theory variant.

Since its introduction in 1922, Zermelo-Fraenkel set theory has become one of the most popular and widely studied set theories. However, not everyone has accepted the Axiom of Choice, leading to the distinction between the ZF and ZFC set theories. We will discuss some of its uses and the debate between ZF and ZFC in later sections.

4 The Banach-Tarski Paradox

In a paper published in 1924, Stefan Banach and Alfred Tarski, provided their formulation of what is now called the "Banach-Tarski Paradox". In their paper, they utilized the controversial AoC, and some other tools, to show that given a sphere in 3-D space, it could be deconstructed and reformed into two spheres that are identical to the original sphere.

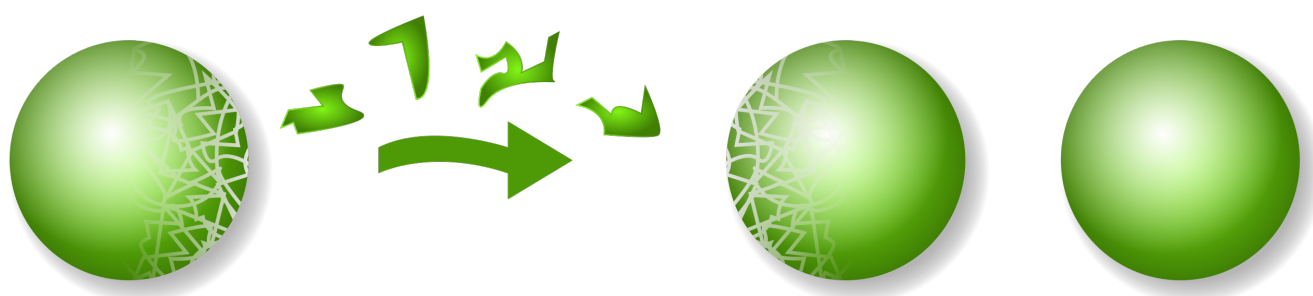


Figure 5.1: Banach-Tarski Paradox Visualization

It should be noted that this paradox does not actually disprove the AoC, and simply calls into question the validity of the tools used to formulate this paradox.
Our proof of the Banach-Tarski paradox relies on some concepts such as non-measurable sets and free groups. There are rigorous definitions for these concepts but we will just stick to a basic understanding of each concept to avoid unnecessary confusion.

Definition 4.1. In 1-D, 2-D, and 3-D spaces we consider the Lebesgue measure as the length, area, and volume of sets. We denote the (Lebesgue) measure of a set X as $\lambda(X)$.

Example 4.2. Here are some examples of measurable sets in $\mathbb{R}^n$ for $n = 1, 2, 3$.

- the interval $[1, 4] \subseteq \mathbb{R}$ has measure 3, because the length of the interval is 3.
- the square defined by $\{(x, y) \in \mathbb{R}^2 : x, y \in [3, 5]\} \subseteq \mathbb{R}^2$ has measure 4, because the area of the square is 4.
- the cube defined by $\{(x, y, z) \in \mathbb{R}^3 : x, y, z \in [0, 3]\} \subseteq \mathbb{R}^3$ has measure 27, because the volume of the cube is 27.

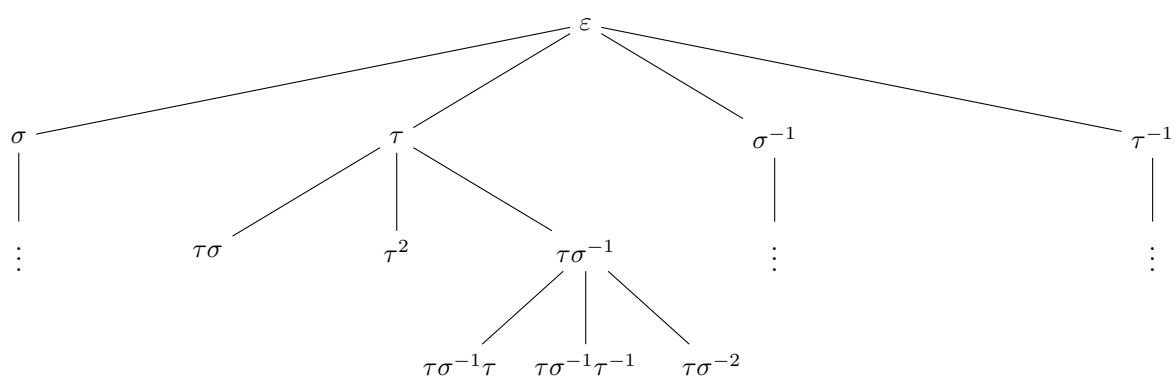
Definition 4.3. We say that a set is <u>non-measurable</u> when we cannot assign a meaningful notion of length, area, or volume to it.

Example 4.4. Here is an example of how to create a non-measurable set in $\mathbb{R}^2$, which actually relies on the Axiom of Choice.
Suppose we have the unit circle given by $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. C is measurable and we can assign $\lambda(C) = 2\pi$ which is the circumference of the circle. Now we define an equivalence relation $\cong$ on C. We say that for any $a, b \in C$, $a \cong b \iff$ there exists a rational rotation that when applied to a, yields b. With this equivalence relation, we have created a partition on C with an uncountably infinite amount of equivalence classes. Each equivalence class has a countably infinite amount of elements, because the class $[p]$ has all the points in C that are a rational rotation away from the point p. Using the Axiom of Choice we know that there is a way to choose a representative for each equivalence class. We Let M denote the set that contains all the representatives for each equivalence class. $\boldsymbol{M}$ **is a non-measurable set.**

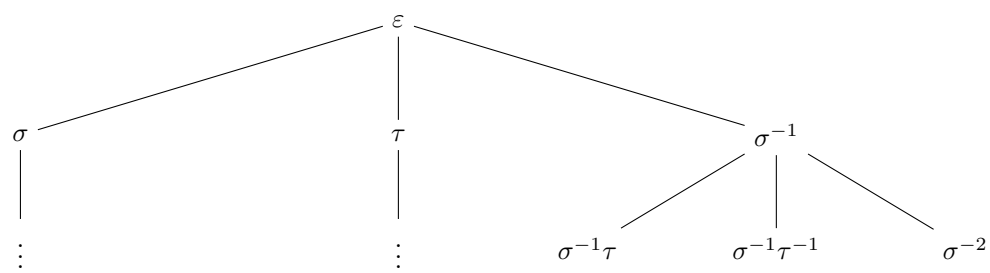
We need the concept of non-measurable sets because in our formulation of the paradox we will partition the sphere (which is measurable) into non-measurable sets that will be manipulated to create two spheres. It is important that the partitioned sets are non-measurable because if they weren't then we'd come to the conclusion that one sphere has the same volume as the total volume of two spheres that are identical to it. However, this would be a contradiction. Thus, since the sets are non-measurable, the contradiction does not occur, which is exactly why the paradox does not disprove the AoC.

Along with the notion of non-measurable sets, we need free groups, specifically free groups generated by two elements.

Definition 4.5. The free group $\mathbf{F}_2$, generated by two elements: σ, τ is the group consisting of all finite products of σ, τ and their inverses σ^{-1}, τ^{-1}. $\mathbf{F}_2$ has a countably infinite amount of elements. Also, $\mathbf{F}_2$ does not contain products where σ or τ is next to its own inverse, i.e. elements like $\sigma^2\tau\tau^{-1}\sigma\tau$ do not exist in $\mathbf{F}_2$, instead the element would be $\sigma^3\tau$. Lastly, $\mathbf{F}_2$ contains an identity element (the empty product) denoted by ε. We can visualize $\mathbf{F}_2$ by expressing its elements in a tree like so:



With this tree diagram its easy to see how we can partition $\mathbf{F}_2$. If we let $W(x)$ denote all the elements in $\mathbf{F}_2$ that begin with x then we can write $\mathbf{F}_2 = \{\varepsilon\} \cup W(\sigma) \cup W(\tau) \cup W(\sigma^{-1}) \cup W(\tau^{-1})$ We notice that the subtree with τ as the root represents the subset $W(\tau)$. The coset $\tau^{-1}W(\tau)$ yields the tree below:



We can see from this tree that $\tau^{-1}W(\tau) = \mathbf{F}_2/W(\tau^{-1})$. Through similar logic we find that $\sigma^{-1}W(\sigma) = \mathbf{F}_2/W(\sigma^{-1})$. We can rewrite these to describe $\mathbf{F}_2$ like so:

$$\mathbf{F}_2 = \tau^{-1}W(\tau) \cup W(\tau^{-1})$$
$$\mathbf{F}_2 = \sigma^{-1}W(\sigma) \cup W(\sigma^{-1})$$

This deconstruction and reconstruction of $\mathbf{F}_2$ is called a *paradoxical deconstruction*, and it will be a very important part of our proof of the Banach-Tarski Paradox. This is called a paradoxical deconstruction because we partitioned $\mathbf{F}_2$ into 5 disjoint sets, and applied some changes to them to reconstruct two copies of the original free group, where no copy uses a set that is in the other copy.

In our proof of the Banach-Tarski Paradox, we will come up with a paradoxical deconstruction of a solid 3-D sphere. It should be noted that the following proof should not technically be called a proof because it will skip over many steps for the sake of simplicity. It should be seen more like a rough outline. We encourage you to read over a mathematically rigorous proof of the Banach-Tarski Paradox if you find the following outline fascinating.

Theorem 4.6. ***Banach-Tarski Paradox.*** *Given a solid sphere in 3-D space, there exists a decomposition of the sphere into a finite number of disjoint subsets, which can then be reconstructed to yield two identical copies of the original sphere. Indeed, the reassembly process involves only moving the pieces around and rotating them without changing their shape.*

Proof. The first step of our proof is to deconstruct a solid sphere in 3-D space into a finite amount of non-measurable sets. We will consider the sphere S with radius 1 centered at the origin $(0, 0, 0)$. We start by defining a way to label points on the surface of the sphere. We pick some starting point on the surface of the sphere and we say that applying x to the point rotates it $\arccos(\frac{1}{3})$ radians around the x-axis. Similarly, we say that applying z to the point rotates it $\arccos(\frac{1}{3})$ radians around the z-axis. We use $\arccos(\frac{1}{3})$ because any sequence of rotations will yield a point different from the starting point. However, there are actually some points where this is not true for every sequence of rotations, for the sake of simplicity we will ignore such points and take for granted that these points are accounted for in the both of the final two spheres.

We can create a free group, $G(x, z)$, generated by x, z where x^{-1}, z^{-1} do the opposite rotations of x, z respectively. $G(x, z)$ has a countably infinite amount of elements, which means there is a countably infinite amount of ways to apply rotations to some point on the surface of the sphere. So, if we pick some starting point on the sphere, we can apply every sequence of rotations to it and achieve a countably infinite amount of points on the surface of the sphere. Now, let us put each of these points into a set based on the last rotation performed in the sequence. Denote $W(x)$ as all the points that have x as the last rotation, and

similarly for $W(z), W(x^{-1}), W(z^{-1})$.

This is good, as we have partitioned some of the sphere into four parts. However, we want to label every point on the sphere, and get a partition of all the points on the sphere. We define an equivalence relation on the set with all points on the sphere: for any points p_1, p_2 on the surface of the sphere, p_1 is related to p_2 if and only if p_2 is the result of performing a sequence of rotations from $G(x, z)$ on p_1. This equivalence relation gives us an uncountably infinite amount of equivalence classes, and each class $[p]$ contains a countably infinite amount of points, since $G(x, z)$ is countably infinite. By the Axiom of Choice we know that there is a way to choose a representative from each equivalence class, so we create a new set E, containing the representative from each equivalence class.

Now, we that we have the set E which is a non-measurable set, we will find a way to apply rotations from $G(x, z)$ to all the points in E to generate disjoint partitions of the entire sphere.

The set $W(x)$ denotes all sequences of rotations that end with x, so the set $W(x)E$ is the set with all rotations that end with x applied to every point in E. $W(x)E$ is a non-measurable set because E is non measurable. We have that the set of all points on the surface of the sphere can be written as $S = W(x)E \cup W(z)E \cup W(x^{-1})E \cup W(z^{-1})E$. Notice that the set $x^{-1}W(x)E$ is actually the same as $S/W(x^{-1})E$, because of the nature of the free group $G(x, z)$. Finally, we have our paradoxical deconstruction of S, which is:

$$S = x^{-1}W(x)E \cup W(x^{-1})E$$
$$S = z^{-1}W(z)E \cup W(z^{-1})E$$

We are actually almost done, since this deconstruction is for just the surface of the sphere, we need to define sets that include the interior points of the sphere as well. We define, $\mathbf{S}$ as the set of all points in the solid sphere including the interior and surface points. Also, define $\mathbf{E}$ as the set of all points in E along with the points inside the sphere that start at the origin and go to a point in E, i.e. the lines that start at $(0, 0, 0)$ and end at p where $p \in E$. With these new sets $\mathbf{S}$ and $\mathbf{E}$ we have a similar paradoxical decomposition:

$$\mathbf{S} = x^{-1}W(x)\mathbf{E} \cup W(x^{-1})\mathbf{E}$$
$$\mathbf{S} = z^{-1}W(z)\mathbf{E} \cup W(z^{-1})\mathbf{E}$$

□

5 Conclusion - The Burden of Choice

The Banach-Tarski paradox helped fuel more doubts about the validity of the Axiom of Choice. This compounded with other criticisms, like that of its idealist outlook: it simply claims that some choice function exists without detailing any other aspects of the function, how many such functions exist, or how one can construct a choice function for each set. In fact, none of the other axioms in Zermelo-Fraenkel set theory can prove or disprove the Axiom of Choice. These factors have led to disagreements which split ZF set theory in two.

As previously discussed, there are two versions of Zermelo-Fraenkel set theory, differing in their decision to include or exclude the axiom of choice - ZF and ZFC. In ZFC, the axiom of choice directly leads to the well-ordering theorem, or that every set can be well ordered. However, this implies that there must be a first (or "least") element in every set. This theorem may cause some issues in certain sets - for example, the sets of real and complex numbers. In ZFC we also have strange results like the Banach-Tarski paradox, which arises from the existence of non-measurable sets - a result of the AoC.

On the other hand, the Axiom of Choice clearly has its uses, as we have shown in the proof of the well-ordering theorem. In addition, the Axiom of Choice (or a statement equivalent to it) has been used in a variety of important proofs, such as:

- A countably infinite union of countably infinite sets is countably infinite.
- The existence of non-Lebesgue measurable subsets of $\mathbb{R}^n$ (as discussed in section 4).
- The cartesian product of a collection of non-empty sets is non-empty.
- Every vector space has a basis.

In the end, the acceptance of the Axiom of Choice comes down to personal preference - and one may find it convenient to accept it under certain conditions and reject it during others. Unfortunately, there still exists no concrete way to prove or disprove the AoC, so its validity - while largely accepted nowadays - remains indeterminate.

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